RESEARCH REPORT · ITERATION 2 · 27 OCT 2026

CV EL research iteration 2

The research no longer waits on ADNI. It now runs on two public datasets: HAM10000 (skin lesions with patient details) and VisA (industrial defects). The lead result is that patient details (age, sex, body site) add clear value to a weak image model but almost none to strong ones, and this holds after controlling for a strong shortcut in the data. It's preliminary: one dataset, frozen image models only, and no novelty claim yet.

Scope and data

DATASET	LICENCE	USED FOR	KEY PROPERTIES
HAM10000 (Harvard Dataverse)	CC BY-NC 4.0	Tracks A and D	10,015 images, 7,470 lesions; age, sex, body site; expert lesion masks
VisA (Amazon)	CC BY 4.0	Track B	12 product categories; official normal-only training split; defect masks
OASIS, ADNI	Data-use agreements	Not used	Parked

- Every HAM10000 split keeps all photos of the same lesion together, because 45% of images come from lesions photographed more than once.
- The main analysis (E10) was **pre-registered**: the predictions were written down before any model was fitted.
- The original notebook, model and paper are unchanged (hashes verified). All 38 tests pass.

Summary of experiments

ID	QUESTION	ANSWER	DATA
E09	Can a capacity curve separate information that acts through the image from information that acts directly?	**Yes. Falling curve = acts through the image; rising = acts directly**	Synthetic

ID	QUESTION	ANSWER	DATA
E10a	Is the HAM10000 metadata safe to use?	Two leaks found and handled: the diagnosis-method field and the image source	Real
E10	Do patient details add less as the image model improves?	**Yes: 0.0186 → about 0.002; 3 of 4 predictions held**	Real
E10b	Is that just a shortcut via image source?	**No. It holds with source controlled; paired test significant**	Real
E10c	Does each variable fade in step with how readable it is from the image?	Inconclusive: all three fade together	Real
E12, E12b	Does the classifier look at the lesion? Are misses due to attention elsewhere?	Evidence sits on the lesion; misses are mostly small lesions	Real
E11	Does whitening features help defect detection?	Yes (0.900 → 0.925), but this reproduces a published method	Real

1. Simulation: what the curve should look like (E09)

In a simulation where the true answer is known, a ladder of image models of increasing quality was combined with an auxiliary variable, as patient age would be.

HOW THE AUXILIARY VARIABLE ACTS	ADDED VALUE, WEAKEST → STRONGEST MODEL	READABLE FROM THE IMAGE (R^2)
Through the image	0.0024 → ~0 (fades out)	0.10 → 0.58
Directly	0.029 → 0.044 (grows)	~0
Both	0.0082 → 0.0016 (falls to a floor)	0.11 → 0.58

The growing direct effect wasn't predicted. It's a known property of logistic models (non-collapsibility). It makes the **direction** of the curve the diagnostic. The derivation is in `research/iteration2/MATH_NOTES.md`.

2. The main result: patient details vs image-model strength (E10)

Task: melanoma vs all other lesions on HAM10000. Five image models of increasing strength were each scored with and without patient details. "Added value" is how much the patient details reduce prediction error (log-loss).

Added value of patient details for each image model (seed 0, 95% lesion-bootstrap intervals)

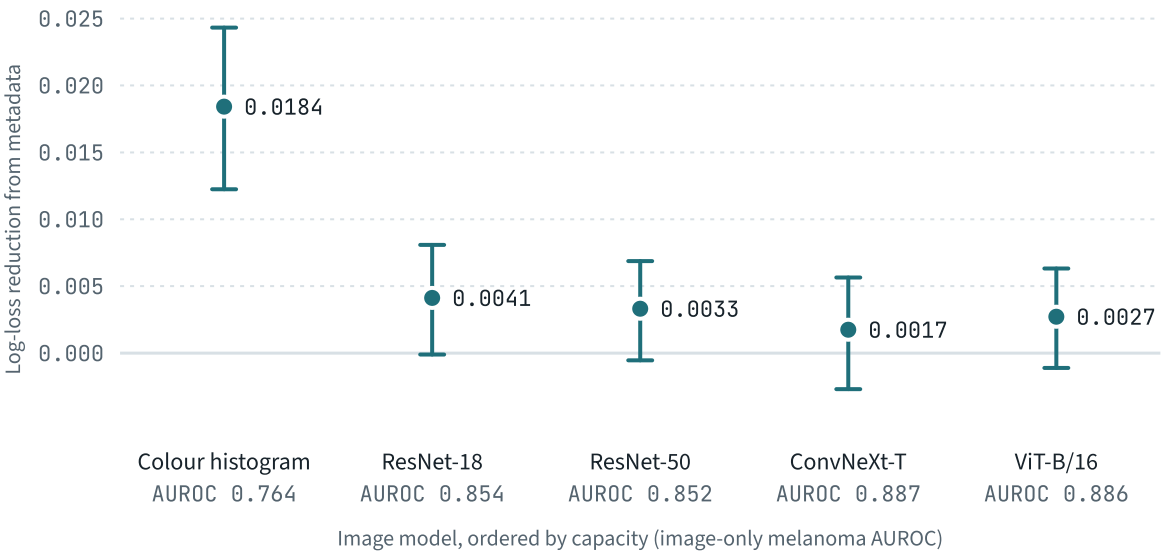


IMAGE MODEL	IMAGE-ONLY AUROC	ADDED VALUE (MEAN OF 3 RUNS)	AGE READABLE FROM IMAGE (R^2)
Colour histogram	0.765	**0.0186**	0.06
ResNet-18	0.853	0.0037	0.28
ResNet-50	0.851	0.0030	0.32
ConvNeXt-T	0.889	0.0017	0.39
ViT-B/16	0.887	0.0022	0.37

Pre-registered predictions:

- Patient details help the weak model: **held** in all 3 runs.
- Value falls as the model improves: **held**, confirmed by paired tests in E10b.
- Some value remains for strong models: **not supported**; their intervals include zero.
- The fall tracks how readable the details are from the image: **held**, qualitatively.

3. Ruling out the shortcut (E10a, E10b)

The image source (which clinic or camera) predicts melanoma on its own (AUROC 0.73), and deep features identify the source almost perfectly (AUROC 0.99). So the deep models' advantage could be partly source detection. Three checks:

CHECK	RESULT
Added value with source given to both models	Still falls: 0.0153 → 0.0047 → 0.0040 → 0.0022 → 0.0032
Melanoma AUROC within a single source	Still rises with model strength: 0.61 → 0.84 and 0.56 → 0.81
Paired test, ResNet-18 vs ConvNeXt (same images)	**Significant: 0.0024 [0.0007, 0.0041]; with source controlled, 0.0026 [0.0009, 0.0040]**

These checks are exploratory: one run, 14 comparisons, no correction for multiple testing. Among the deep models the absolute differences are small, about 1% of prediction error.

Mechanism (E10c): inconclusive. Age, sex and site all lost value together and all became readable from the images. That includes sex (AUROC 0.77), contrary to my expectation. So this test couldn't single out why the value fades.

4. Where the classifier looks (E12, E12b)

The classifier's evidence maps were compared with the experts' lesion outlines, using the validated mask-shift test from iteration 1 in a new exact form.

FINDING	EVIDENCE
Melanoma evidence sits on the lesion	84% of images; median 1.49× more evidence on the lesion than its area share
Pixel shuffling over-declares significance	82% vs 37% significant (valid test); 30% vs 10% in E12
Missed melanomas are mostly small lesions	Detected 52% of image area vs missed 37% ($p = 3\times10^{-13}$); within size groups, attention placement barely differs

A pitfall I hit and corrected. My first version mapped the evidence for whichever class was *predicted*. For "not melanoma" predictions that evidence sits on the surrounding skin, so errors (mostly missed melanomas)

looked "off-lesion" purely because of how the maps were built. Using the melanoma evidence map for every image removed the effect.

5. Industrial defects (E11)

On all 12 VisA categories, using the official split, whitening the features before nearest-neighbour matching raised mean image AUROC from 0.900 to 0.925 and pixel AUROC from 0.976 to 0.983. It was better in 8 of 12 categories (Wilcoxon p = 0.052). The largest gain was on macaroni2, from 0.62 to 0.77.

This only reproduces a published 2026 method (Mahalanobis PatchCore), so the track is closed. Lesson: the first category tested (candle) pointed the other way, so conclusions from a single category would have been wrong.

6. M5 performance

WORKLOAD	MEASURED
Image embedding (ResNet-18 / ResNet-50 / ConvNeXt-T / ViT-B)	638 / 195 / 171 / 61 images per second
Whole pre-registered E10 analysis	2.4 minutes
VisA defect detection, per category	about 2.4 minutes

With frozen image models, compute is not the bottleneck; downloads and analysis are. Fine-tuning (next iteration) will be the first compute-heavy step. No energy figures were measured.

Novelty status

ITEM	STATUS
Patient-detail value fades with image-model strength (HAM10000)	Supported on one dataset. No prior controlled study found in 4 searches, but one possible counter-example paper (CVPR 2025 workshop) must be read in full
Curve direction separates "through the image" from "direct"	Derived and supported in simulation; builds on a known statistical property
Valid significance test for saliency vs anatomy, with exact FFT form	Idea known from brain mapping; applying it to AI explanations is possibly new (unverified)

ITEM	STATUS
Missed melanomas are mostly small lesions	Supported for one model; no size-stratified study found yet
Feature whitening for defect detection	Known (reproduced)

Decision

Lead direction: the patient-details-vs-model-strength question (Track A). It has the best mix of real-data evidence, a careful design and a plausible gap in the literature. It also keeps the original project's idea, "does extra information help beyond imaging?", on data we can actually use. The saliency-testing work (Track D) is a strong supporting component. The defect-detection track is closed.

Before this becomes a paper:

1. Repeat the analysis unchanged on a second skin dataset (PAD-UFES-20 or ISIC 2020), with tests corrected for multiple comparisons.
2. Test fine-tuned image models, not only frozen ones.
3. Read the closest papers in full, including the possible counter-example.

Mistakes and corrections this iteration

- E12's first analysis was confounded by how the evidence maps were built. Fixed in E12b.
- I predicted sex would be hard to read from skin images. It wasn't (AUROC 0.77).
- I nearly closed the defect track after one category; the full run reversed that.
- I briefly overstated that whitening helps most on the hardest categories. The correlation isn't significant ($p = 0.21$).
- Download time estimates were badly off, because the connection speed varied between 0.2 and 7 MB/s.

Files

Everything is in `~/Desktop/CV EL/research/`. Iteration 2 material is in `iteration2/`: the dataset comparison, the pre-registration, the maths notes and the decision report. All experiment results are in `EXPERIMENTS.md`, with scripts in `experiments/` and raw results in `results/`. The downloaded data takes 12 GB in `research/data/`.

Prepared with Claude Code. Results on public research datasets only; not clinical evidence. The iteration-1 report is kept unchanged as the record of that round.